POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2013
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Argue against the "Turbine test" as providing a true test of intelligence. 7

b) A farmer has a goat, a wolf & a cabbage on the west side of a river. He wants to get all of his animals & his cabbage across the river onto the east side. The farmer has a row boat but he only has enough room for himself & one other thing. The wolf will eat the goat if they are left together alone. The goat will eat the cabbage if they are left together alone. How can the farmer get everything on the east side? 8

i. Formulate this puzzle as search
ii. Find a solution for this problem.

a) Solve the following crypto arithmetic problem. 7

```
 FORTY
   TEN
 +TEN
------
 SIXTY
```

b) "Breadth first search is an implementation of queue whereas Depth first search is an implementation of stack". Verify this statement with suitable example. 8

a) What is hill climbing search? Explain the problem associated with this search techniques. 8

b) Explain Alpha-Beta Pruning search. What are the advantages of Alpha-Beta Pruning over Min-max search? 7

a) Briefly explain the limited expressive capabilities of propositional logic. How does the first order logic extend propositional logic to overcome these limitations? 8

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- All the course in the basket weaving department are easy
- BK301 is a basket weaving course

Use resolution to answer the question," What course would Steve like?"

5 a) Define proposition and a predicate. How does first order logic extend propositional logic to overcome its limitations?

b) What is a semantic net? For the statement, "The dog bit the mail carrier." Now, represent this statement (Knowledge) using partitioned semantic nets.

6 a) What is learning? Explain rote learning and induction learning. Why the process of knowledge acquisition so tedious.

b) Why Natural Language understanding is difficult? And also explain its application.

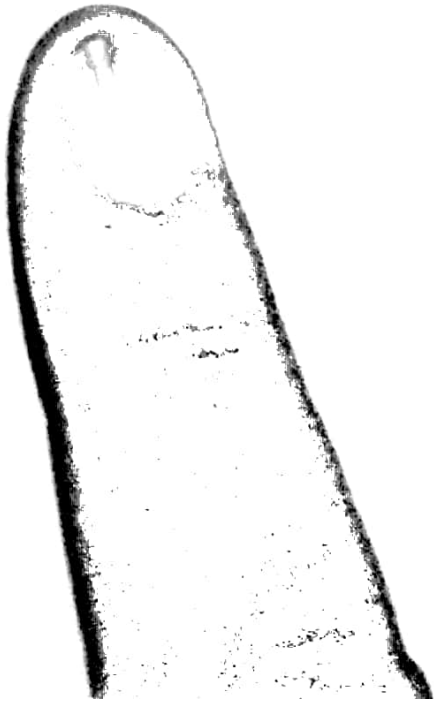
7. Write short notes on: **(Any two)**

a) Statistical reasoning

b) Expert System

c) Mathematics and Artificial Intelligence.

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